Section Overview

- Throughput, Arithmetic Intensity, and the Roofline Model

Driving Question

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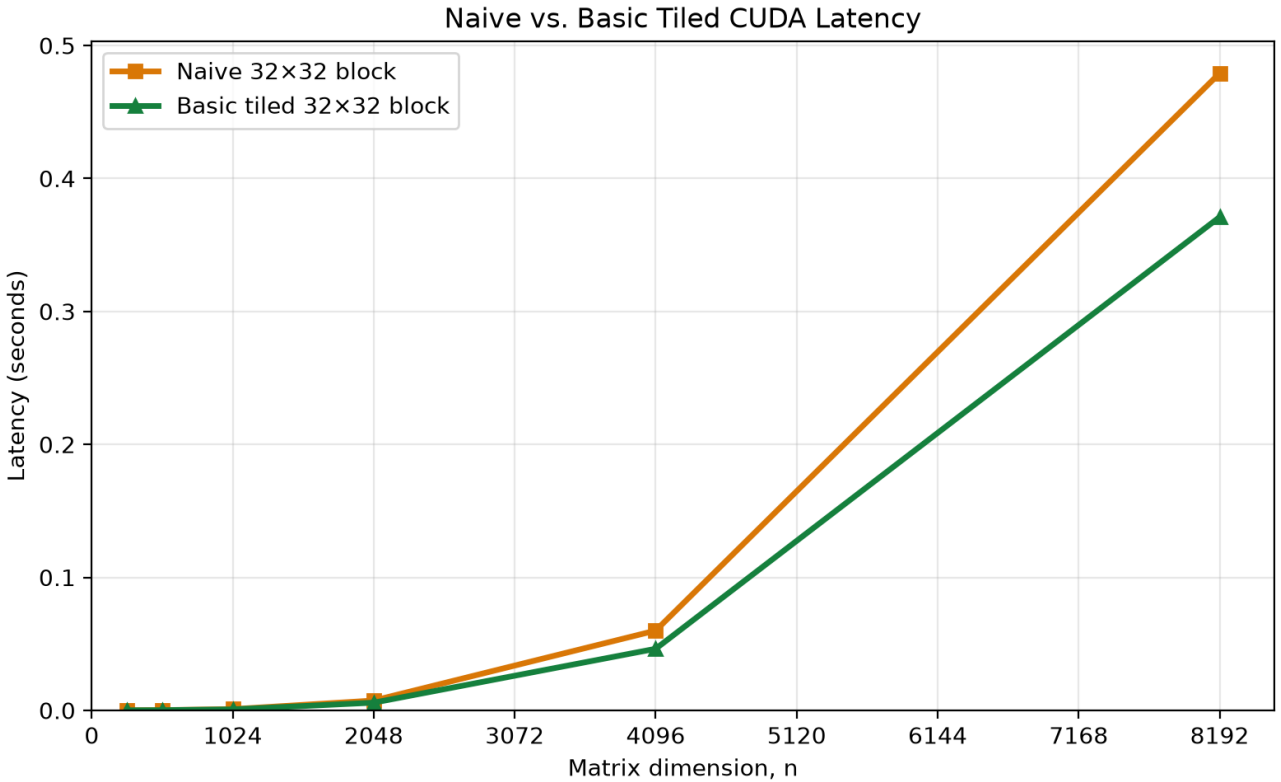
We have been measuring our kernels with **latency**, but is there some other metric that better describes "performance"?

How fast *can* we be?

Given that we can do 35.6×10^{12} floating point operations **per second**, how fast (theoretically) can we do our SGEMM operations?

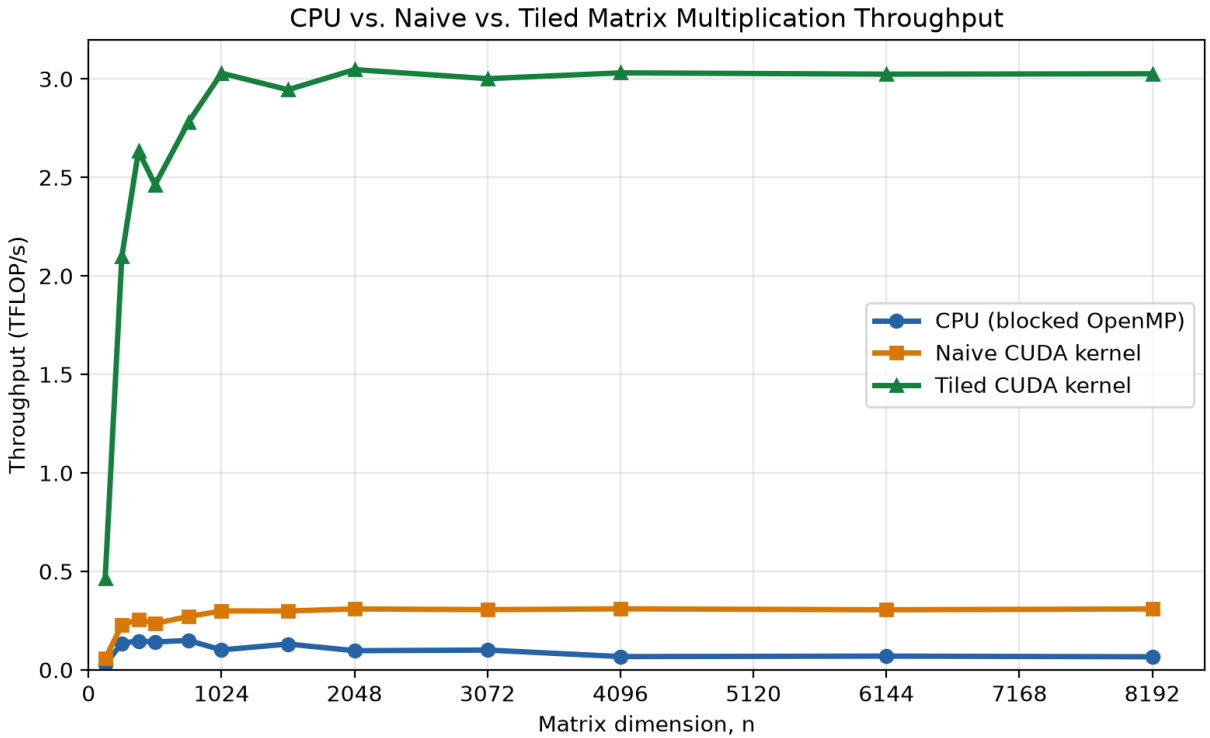
Why latency alone isn't enough

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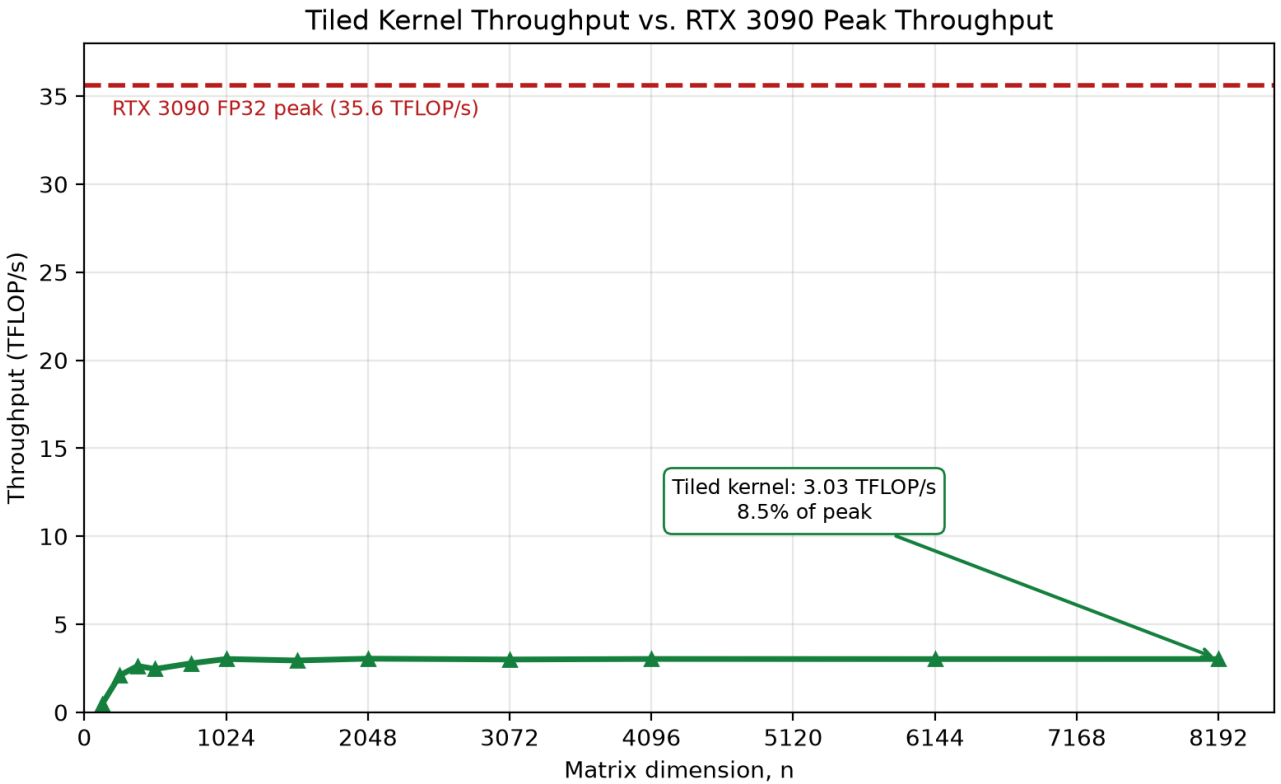
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Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

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Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput.

Characterizing our kernels data use

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1. Throughput characterizes how well our kernel utilizes the hardware
2. One factor that impacts final throughput is how efficiently our kernels use data

Arithmetic Intensity: Characterizing our kernels

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We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

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Example

For $n, k, m = 8192$, our tiled kernel moved a total of 137.7 GB of data from DRAM. With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \times 10^{12} \text{ FLOP}}{137.7 \times 10^{9} \text{ bytes}} \approx \boxed{8.0 \text{ FLOP/byte}}.$$

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Our tiling kernel has AI of 8.0 FLOP/byte, meaning it does 8 floating point operations for every byte of data moved from DRAM.
At $935.8\,\mathrm{GB/s}$ of bandwidth, our **memory roof** is

$$935.8\,\mathrm{GB/s} \times 8.0\ \text{FLOP/byte} \approx 7.49\ \text{TFLOP/s}.$$

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This is much lower than the peak throughput of $35.6\,\mathrm{TFLOP/s}$. **No matter how we optimize this kernel, without increasing AI, we will never achieve greater than 7.49 TFLOP/s**.

Hypothetical Kernel

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$$935.8\,\mathrm{GB/s} \times 16.0\ \mathrm{FLOP/byte} \approx 15.0\ \mathrm{TFLOP/s}.$$

Since $15.0\ \mathrm{TFLOP/s} < 35.6\,\mathrm{TFLOP/s}$, this kernel will still be memory bound.

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AI (FLOP/byte)	Upper bound throughput (TFLOP/s)
8	7.49
16	14.97
32	29.95
36	33.69
38	35.56
40	37.43

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Wait! Our peak throughput is $35.6\,\mathrm{TFLOP/s}$! At AI ≥ 38, we are **compute bound**. Our data can finally keep up with the CUDA Cores!

Ridge Point

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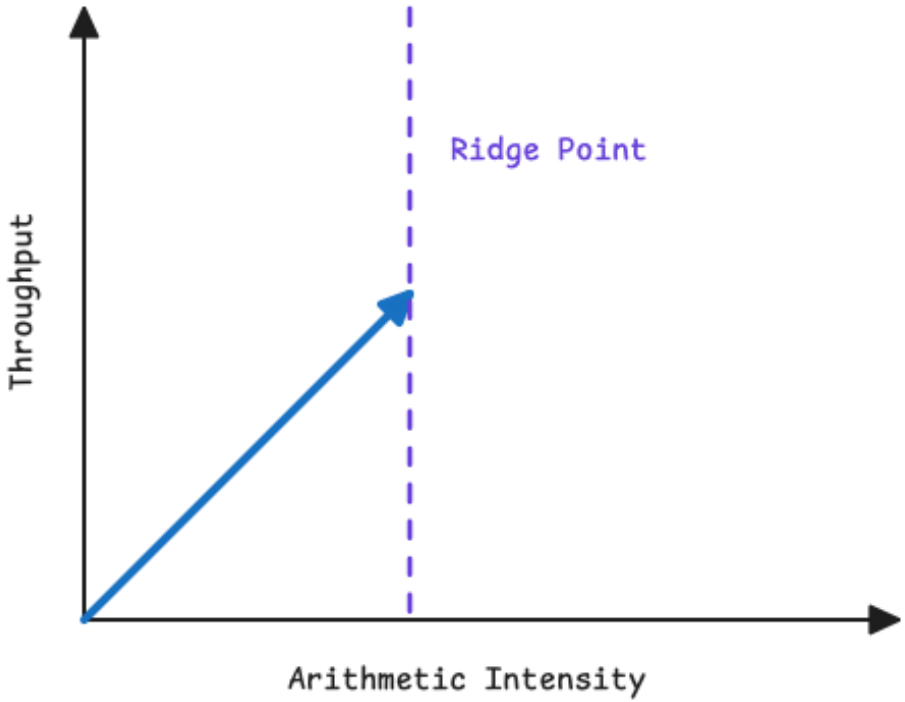
AI $= 38$ FLOP/byte is known as the **ridge point** *RP*, and can be found by solving:

$$\begin{aligned}
\text{Data roof} &= \text{Peak throughput} \\
\text{DRAM Bandwidth} \times \text{RP} &= \text{Peak throughput} \\
\text{RP} &= \frac{\text{Peak throughput}}{\text{DRAM Bandwidth}} \\
\text{RP} &= \frac{35.6\,\mathrm{TFLOP/s}}{935.8\,\mathrm{GB/s}} \approx 38.0 \text{ FLOP/s}
\end{aligned}$$

Roofline Model

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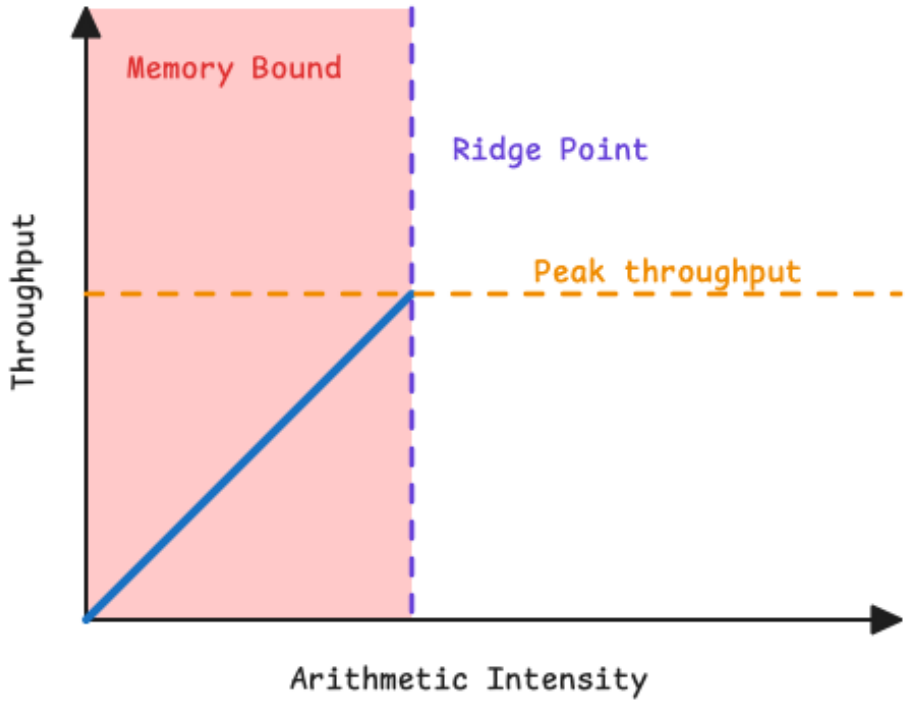
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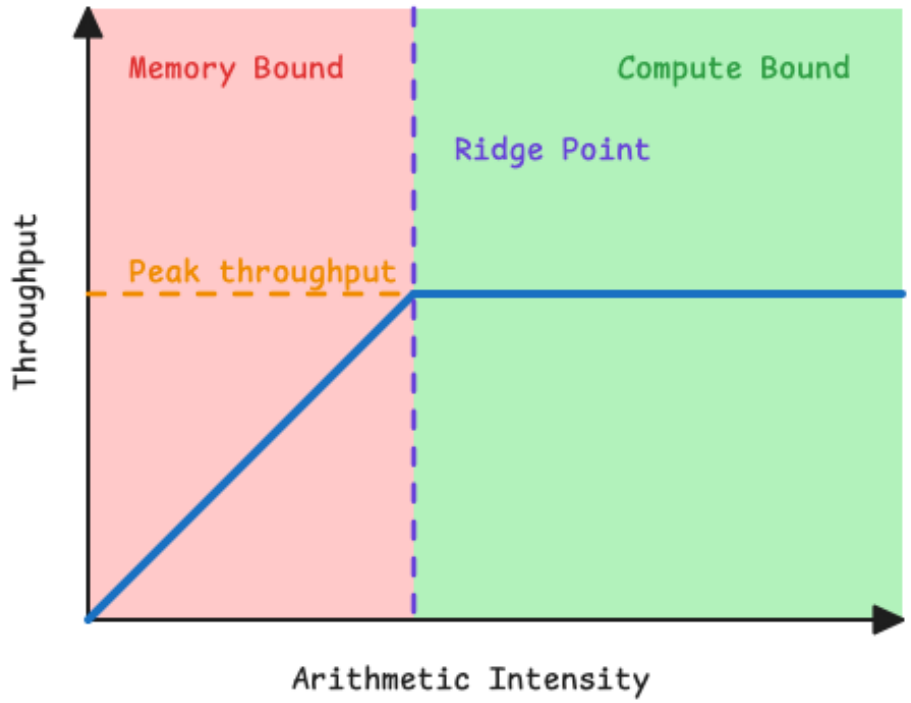
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For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI. Up to the **ridge point**, we are **memory bound**, increasing AI will raise the upper bound of throughput. Past that, we are **compute bound**, increasing AI will not raise the upper bound of throughput.



Roofline Model of our RTX 3090

